

SLA08

Model Selection Manual



CZ SUNLINE

Linear actuator



Dear Customer,

The company upholds the military spirit of "people oriented, honest and trustworthy, innovation and self-improvement, and pursuit of excellence" and the philosophy of "unity, integrity, pragmatism, innovation, efficiency, development and dedication",

returns to the users with the qualified products, wins the market with the technical improvement and explores the markets with the excellent services.

The struggle is the core of corporate culture and the technology is the our development of essence.

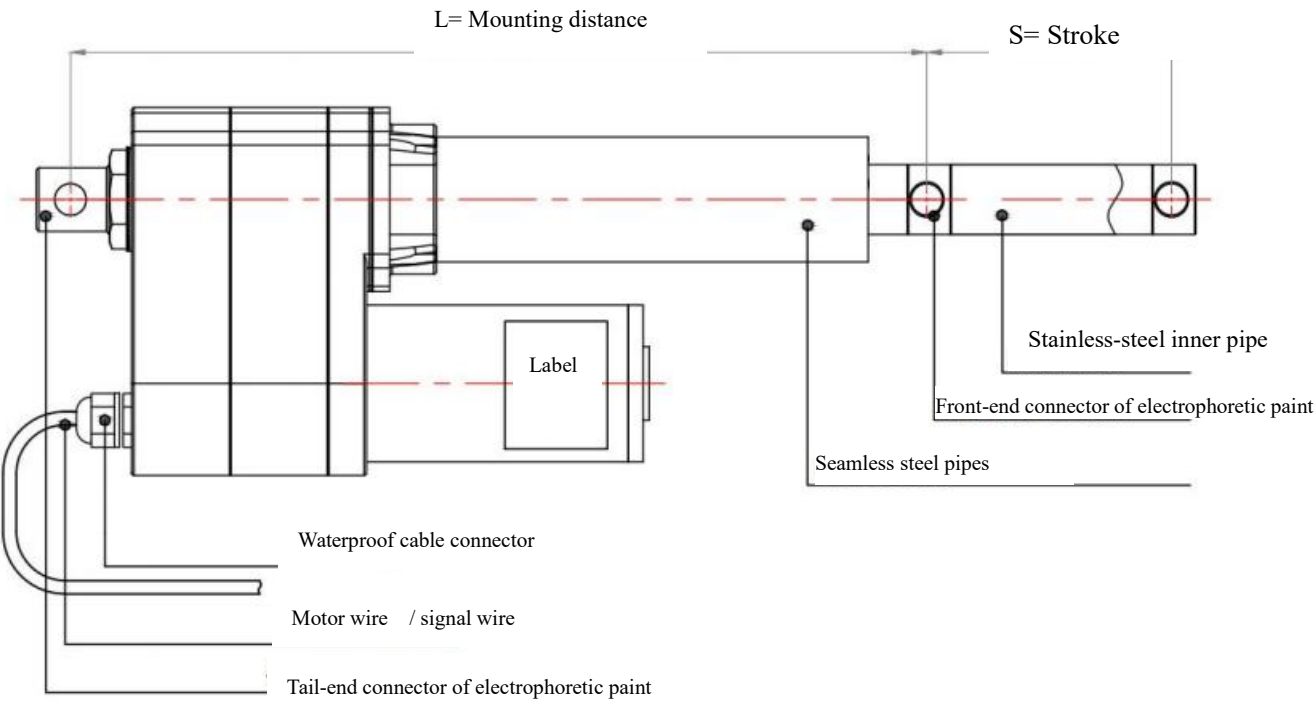
The customer satisfaction is the greatest wish of our employees. We will respect you and achieve the joint development with our sincere.

Please read the model selection manual to have a deeper understanding on the products, ensuring that every putter is the unique qualified product for you.

We also wish that you could put forward the valuable opinions and suggestions and your suggestions will drive our growth.

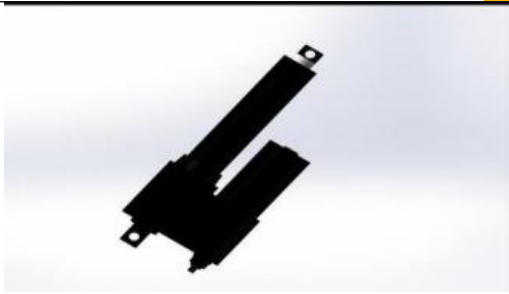
Technical Team

Term definition



Stroke	Maximum moving distance of the push pod, i.e. the fully expanded dimension minus the fully retracted dimension. Customizable.
Mounting distance	Dimensions when the putter is fully retracted, it could be customized.
Front-end connector	More models are available.
Tail-end connector	More models are available.
Mounting hole	0° or 90° (optional)
Dynamic load	Maximum load that the putter could bear under the moving state.
Static load	It refers to the self-locking force, namely the maximum load that the putter could bear under the stationary state.
Protection grade	IP XX: The first X refers to the dust prevention and the second X refers to the water prevention. The larger the number, the better the protection grade. Please inquire [Table 1].
Duty ratio	After work for a minutes, it shall rest for B minutes. The duty ratio is the percentage of a / (a + b). Please inquire [Table 1].
Speed	The no-load speed and full-load speed are negatively correlated with the maximum output force.
Hall sensor	Output voltage pulse and realize displacement measurement by pulse counting. Besides, identify the moving direction of the putter with the waveform phase difference. Please inquire [Table 1] to confirm whether this model supports the Hall sensor.
Potentiometer	The potentiometer is the variable resistor and it could feed back the position according to the resistance variation. Please inquire [Table 1] to confirm whether this model supports the potentiometer.
Manual control position	The optional hand crank could be inserted to replace the electric control and it may drive the device under the emergent conditions. Please inquire [Table 1] to confirm whether this model supports the manual control.

Basic configuration

Shell color	☐ Black grey	☒ Black	☒ ⑤ Other colors			
Screw rod type	☒ Trapezoid screw	☒ Ball screw				
Operating method	☒ Electric control	☐ Electric + manual control				
Application field	☒ Industry	☐ Home Supplies	☐ Medical treatment			
Operating temperature	☐ 5°C - 40°C	☐ -20°C to +65°C	☒ -40°C to +65°C			
Working noise	☐ ≤45dB	☒ ≤65dB	☐ ≤75dB			
Travel range	☒ 50-600mm	☒ 600-1,080mm				
loading range	☐ ≤1,200N	☐ ≤2,000N	☐ ≤7,000N	☒ ≤10,000N	☐ ≤12,000N	
Duty cycle %	☐ 10%	☐ 20%	☒ 25%	☐ 50%	☐ 100%	
Motor type	☒ DC brush	☐ Stepping motor	☐ Brushless motor	☐ Actuating motor		
Overload protection	☒ None	☒ Mechanical clutch	☐ Electronic protection	☒ Thermistor		
Protection grade	☐ IP20	☐ IP43	☐ IP54	☒ IP65	☐ IP66	
Signal output	☒ None	☒ Limit signal	☒ Hall sensor	☒ Potentiometer	☐ Encoder	☐ Magnetic reed switch
Input voltage	☒ 12VDC	☒ 24VDC	☒ 36VDC	☒ 48VDC	☐ 110VAC	☐ 220VAC

☒ Available items for the current product

☐ Other products

[Table 1]

Technical parameters

Please fill in the code:

Transmission code	Maximum dynamic load	Self-locking force	Reduction ratio	Thread pitch	Speed ±10%①② (NM/S)		Maximum stroke ③		
							No limit switch and no potentiometer	Have limit switch without the potentiometer	With limit switch and potentiometer
	(N)	(N)		(mm)	No-load	Full load	(mm)	(mm)	(mm)
A	9,000	11,000	40:1	P3/1	5	3.5	1000	400	400
B	7000	10,000	40:1	P2.5/2	8	6.5	1000	610	610
C	5,000	7,000	20:1	P3/1	10	8.5	1000	400	400
D	4,000	5,500	20:1	P2.5/2	16	13	1000	610	610
E	3,000	4,000	10:1	P3/1	20	16	1000	400	400
F	2000	2,800	10:1	P2.5/2	35	28.5	1000	610	610

[Table 2]

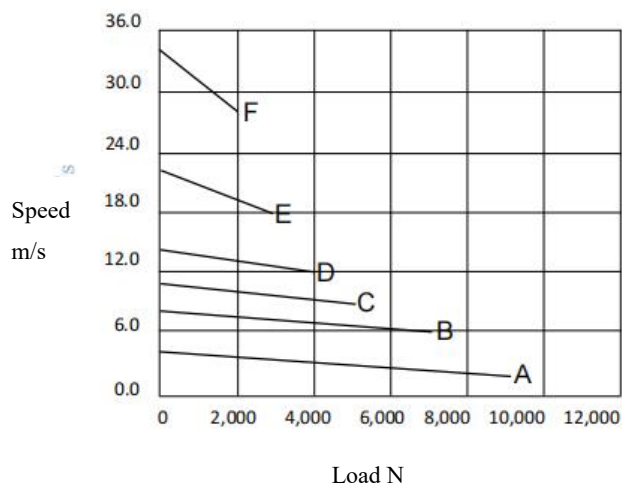
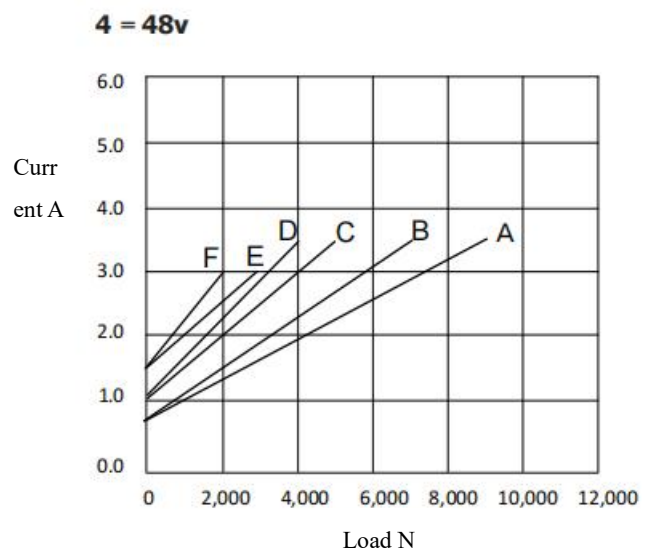
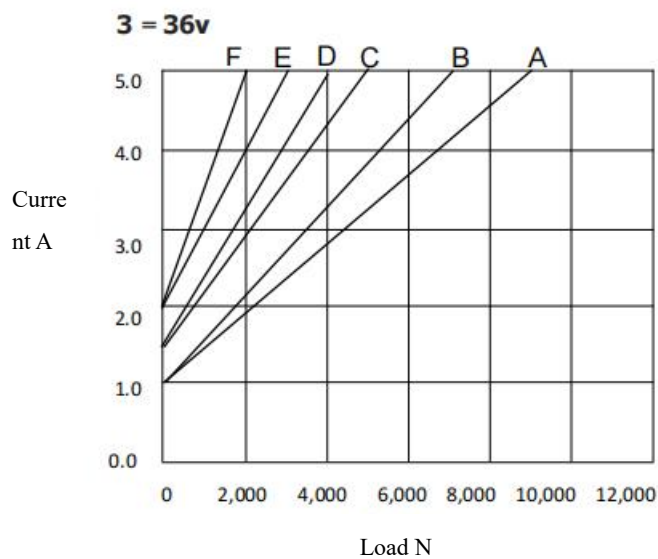
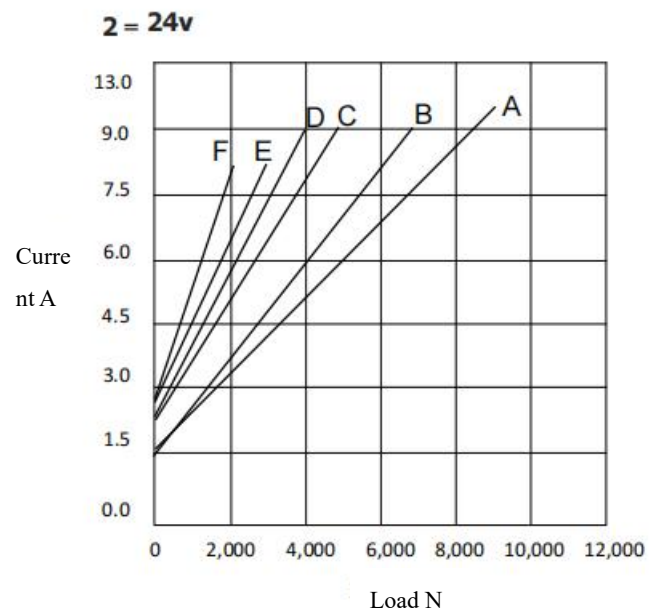
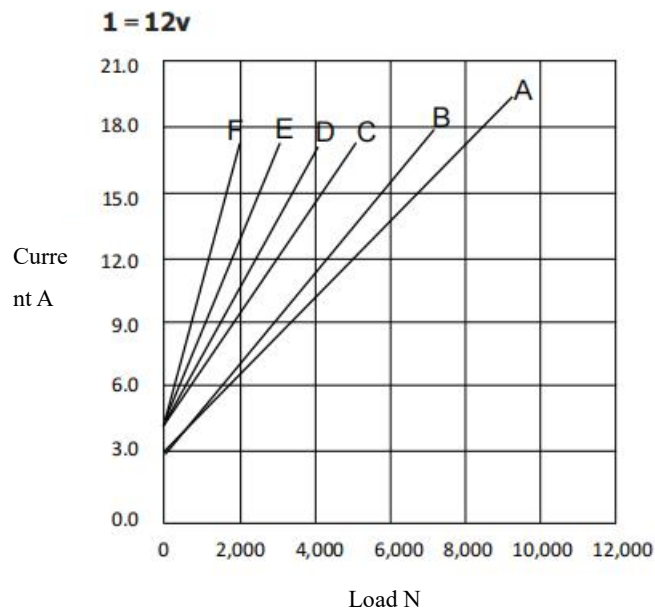
① The data are measured under the ambient temperature (20°C) and the stable voltage.

② When the load is 5,000N, the gear (C) has met the load demands. Of course, you could also select (A) or (B), so that you could obtain the higher load margin, safety coefficients and longer service life.

③ There are more factors affecting 'customizable maximum stroke', such as load, speed, direction of force and frequency of use. Hence, you shall combine with the real application scenario. If the parameters required exceed the available range, please contact the sales engineer for further evaluation.

Relational graph

Please fill in the code:

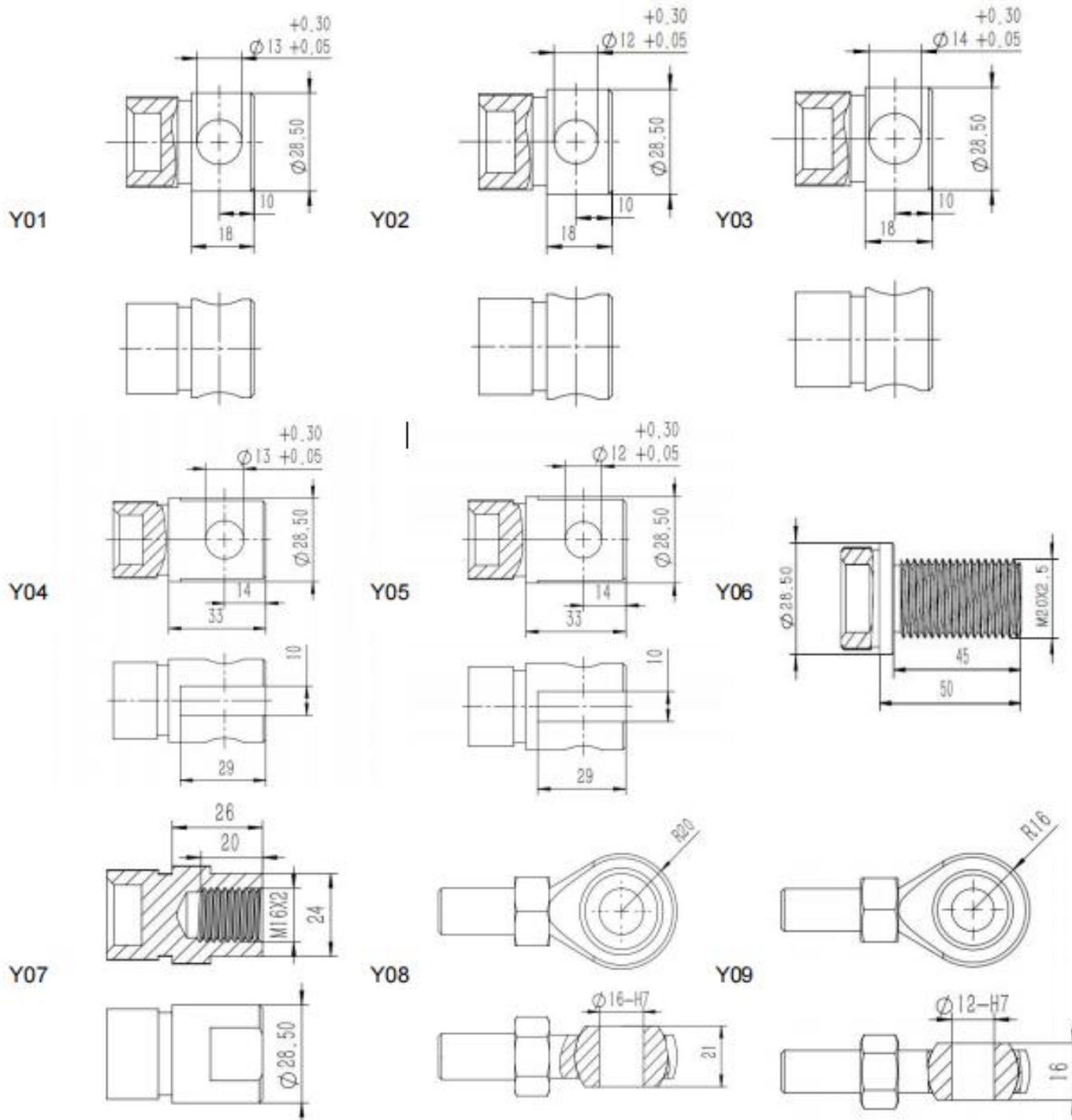


* The data are measured under the ambient temperature (20°C) and the stable voltage.

Front-end connector

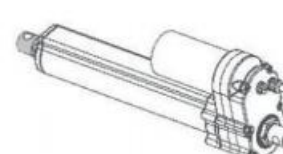
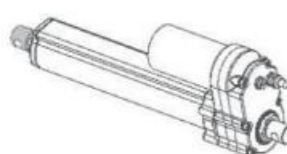
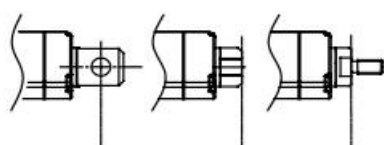
Please fill in the code:

1. More models are available or contact our company for the personalized customization.



2. Definition of mounting distance

3. Hole direction of the front-end connector Please fill in the code:



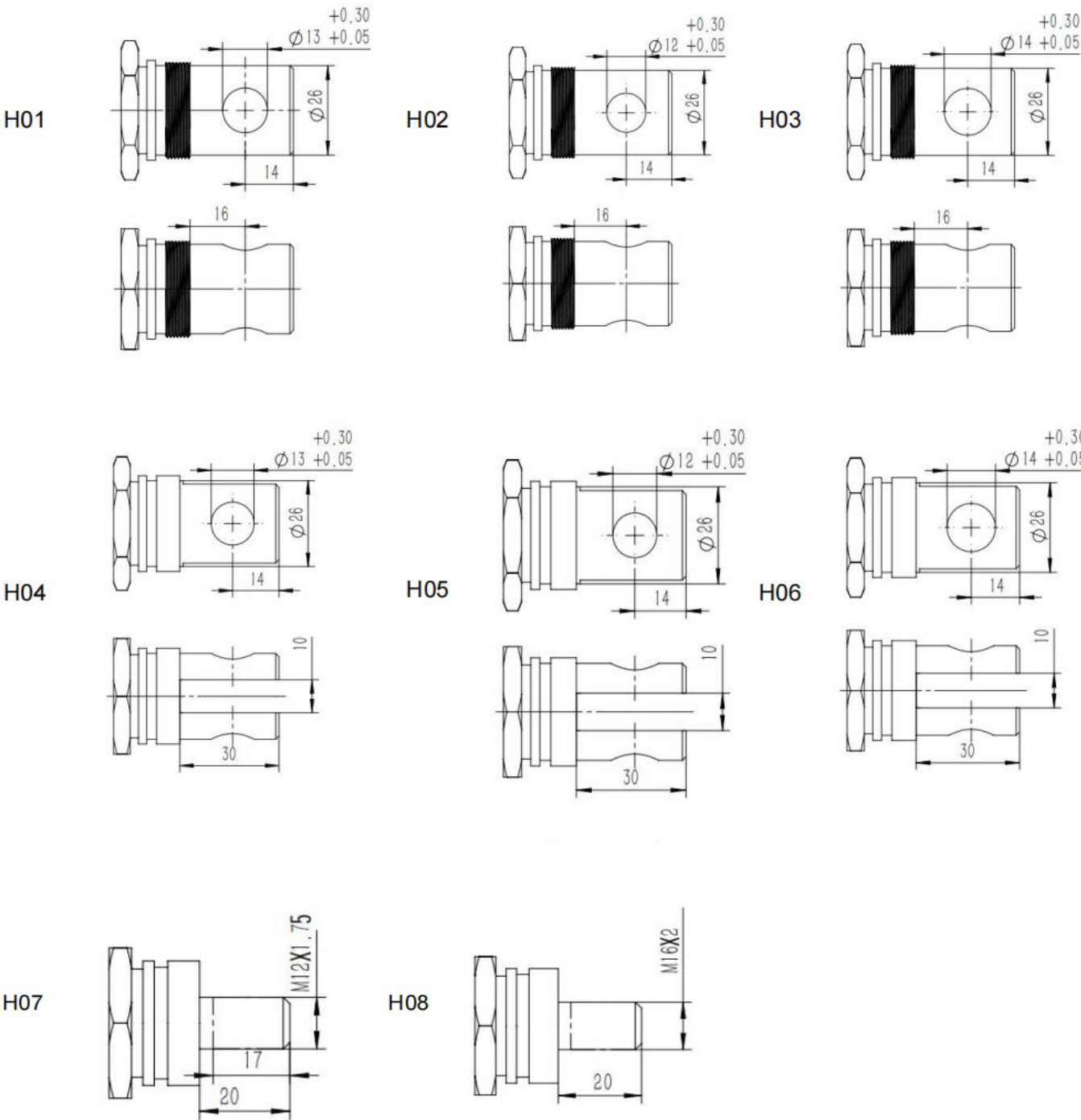
1 = 0°

2 = 90°

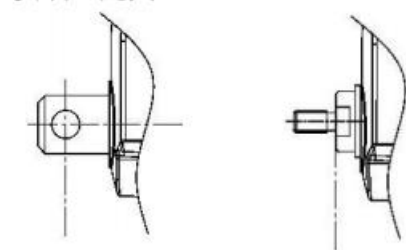
Tail-end connector

Please fill in the code:

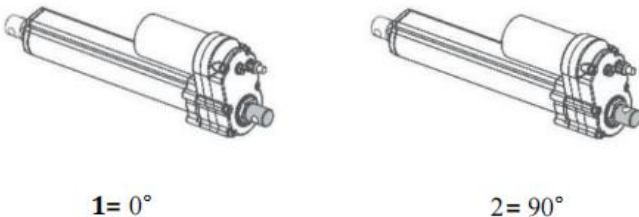
1. More models are available or contact our company for the personalized customization.



2. Definition of mounting distance



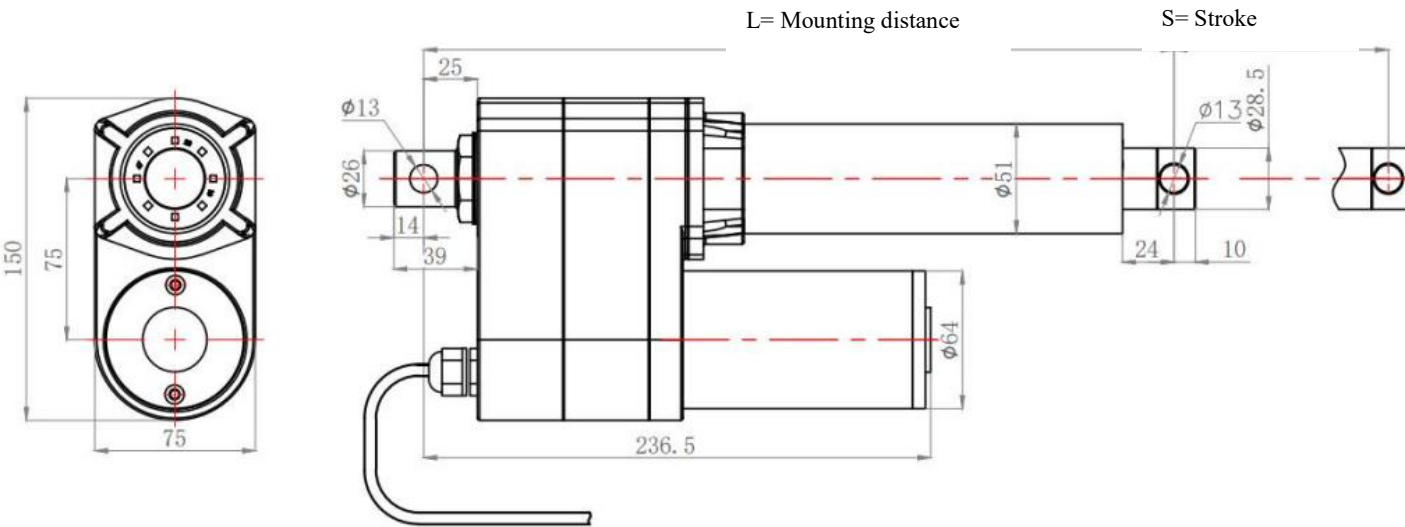
3. Hole direction of the front-end connector Please fill in the code:



Dimension

[Built-in limit switch]

Stroke S=mm, Mounting distance L=mm



A. Front and rear connector VS mounting distance

Front-end connector (skip)	Tail-end connector (skip)
	H01, H02, H03, H04, H05, H06, H07, H08
Y01, Y02, Y03, Y04, Y05, Y06, Y07	$A \geq S + 250\text{mm}$
Y08, Y09	$A \geq S + 300\text{mm}$

[Table 3]

B. Stroke VS mounting distance

Stroke S (mm)	Mounting distance L (mm)
50-399	+0
400-610	+50

[Table 4]

Computational formula of mounting distance

S = Stroke

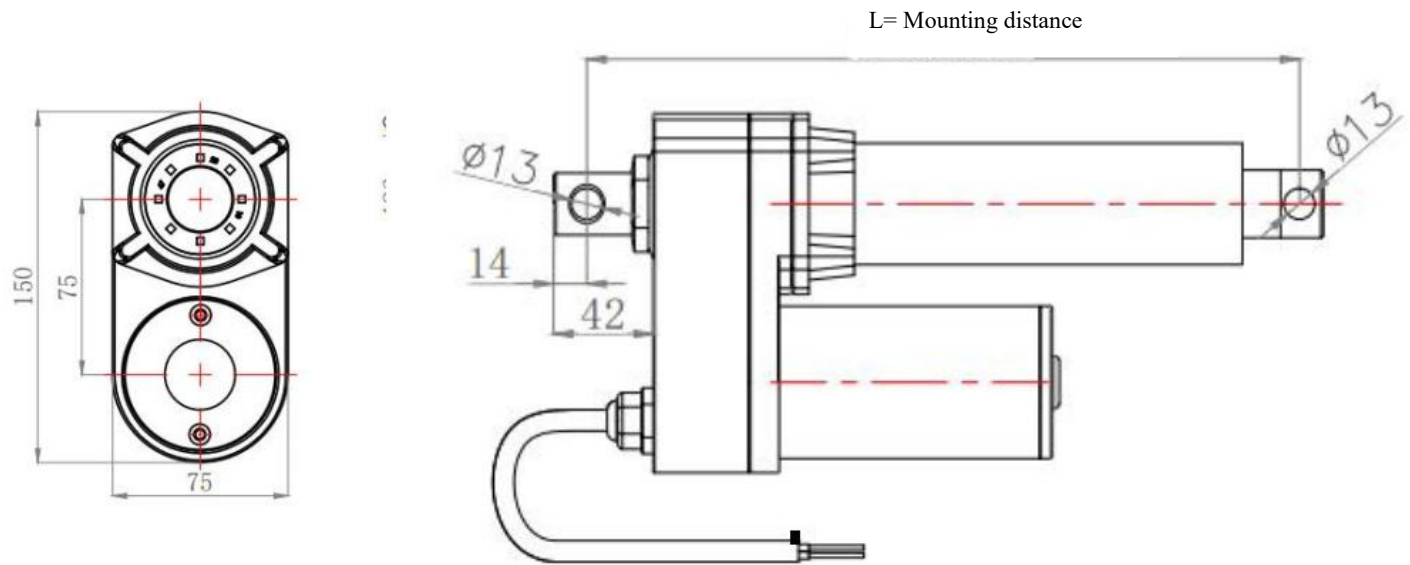
L = Mounting distance

$L \geq A + B$

Dimension

[no limit switch]

Stroke S=mm, Mounting distance L=mm



A. Front and rear connector VS mounting distance

Front-end connector (skip)	Tail-end connector (skip)
	H01, H02, H03, H04, H05, H06, H07, H08
Y01, Y02, Y03, Y04, Y05, Y06, Y07	$A \geq S + 200\text{mm}$
Y08, Y09	$A \geq S + 250\text{mm}$

[Table 5]

B. Stroke VS mounting distance

Stroke S (mm)	Mounting distance L (mm)
50-399	+0
≥ 400	+50

[Table 6]

Computational formula of mounting distance

S = Stroke

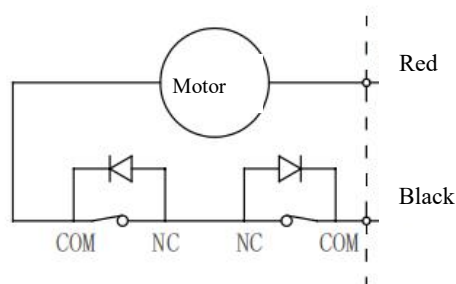
L = Mounting distance

$L \geq A + B$

Signal feedback

0. The built-in limit switch has no signal feedback.

This model includes the upper limit and lower limit switch by default. When the putter could be fully extended or fully retracted, the limit switch automatically cuts off the power supply of the motor.



Wiring mode of telescopic action		
Wire color	Black	Red
Extend	-	+
Contract	+	-

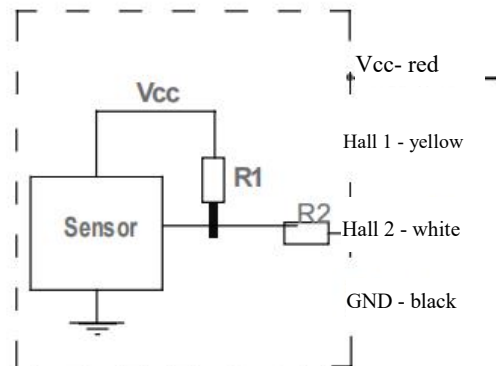
[Table 7]

1. Normally open signal out of upper and lower limit

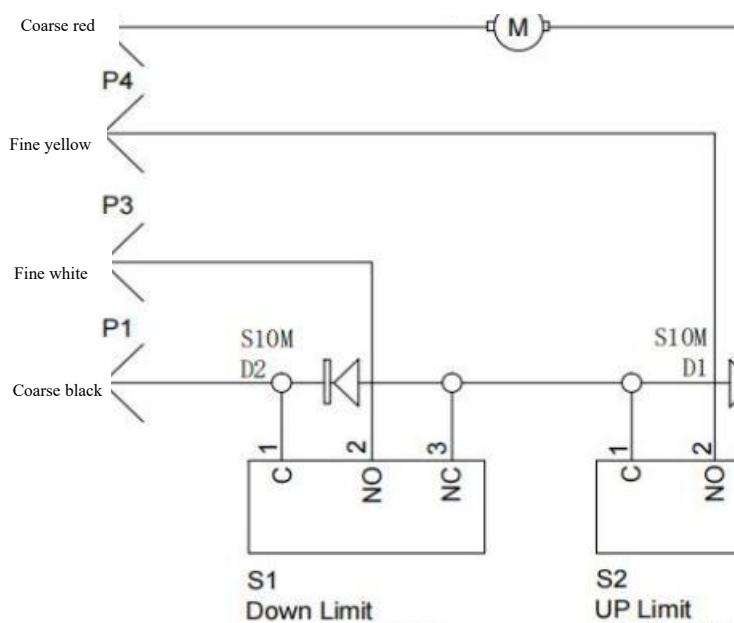
The model is passive by default. When the putter could be fully extended or fully retracted, the limit switch automatically cuts off the power supply. As shown in the figure below

2. Hall sensor (double hall in standard configuration)

Schematic diagram of internal wire and corresponding diagram of external wire color of Hall signal



At the front limit, connect the fine yellow and coarse black, suspend the fine white.
At the rear limit, connect the fine white and coarse black, suspend the fine yellow.
At the middle position, suspend the fine yellow and fine white.



Rear limit (extracted to the minimum length)

Front limit (extended to the maximum length)

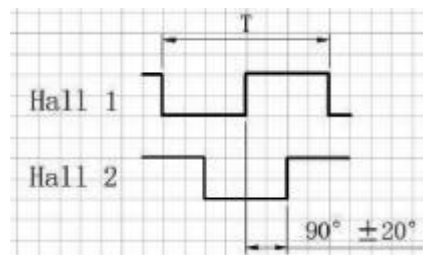
Hall Signal Feedback Form

Transmission code	Pole options	Pulse equivalent *pulse/mm
		Four-pair poles
A	b. Four-pair poles (standard configuration)	50.5
B		32.0
C		25.2
D		16.0
E		12.6
F		8.0

* The number of the pulse signals corresponding to the 1mm stroke of the single Hall.

[Table 8]

Output Waveform Diagram of Hall Signal



Applicable scope V=5~15V

3. Potentiometer

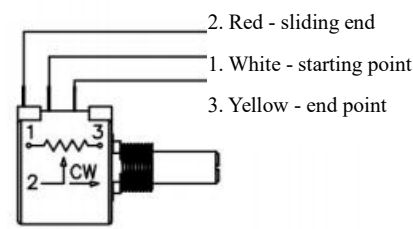
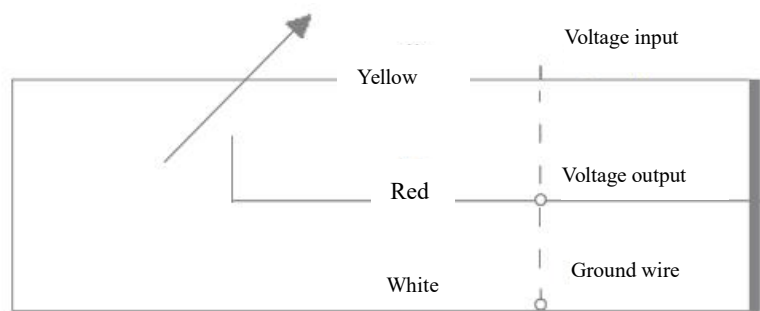
Configuration scope of the potentiometer

Transmission parameter code	Maximum stroke supported	Resistance value of 1mm stroke
A,C,E	400mm	0.0250K Ω
B,D,F	610mm	0.0159K Ω

Note: Starting value of potentiometer resistance $0.5\pm0.1K\Omega$

[Table 9]

Schematic Diagram of Potentiometer



After the putter is extended, the white and red resistance value will become greater.

	Voltage	1 = 12VDC 2 = 24VDC 3 = 36VDC 4 = 48VDC
	Transmission code	For more details, please refer to [Table 2]
	Stroke (mm)	If it exceeds the optional range, the practical application scenario may be combined for further evaluation.
	Mounting distance (mm)	See [Table 3]and [Table 4] or [Table 5] and [Table 6] for details.
	Front-end connector	Y01-Y09 or YX= customized
	Tail-end connector	H01-H08 or HX=customization
	Hole direction of the connector	Front end 1 = 90° 2 = 0° Tail end 1 = 90° 2 = 0°
	Signal feedback	0 = no signal feedback 1 = normally-open limit signal 2 = Hall sensor 3= potentiometer
	Cable length	1 = 500mm 2 = 8,000mm 3 = 1,000mm X = Customized
	Cable connector	0 = Tinned bare wire, 1 = match with KZ series controller X = customization
Application scenarios	Use frequency	Daily cycle number
	Usage mode	Please describe
		Company name
	Your contact information	Contact
		Tel.
		E-mail

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